

JAHANVI KALRA

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EDUCATION

A. James Clark School of Engineering, University of Maryland

Master of Science, Telecommunication, GPA: 3.9/4.0

College Park, MD

Aug 2017-May 2019

Mumbai University - Watumull Institute of Electrical Engineering and Computer Technology

Bachelor of Engineering, Electronics and Telecommunication, GPA: 7.8/10.0

Maharashtra, India

May 2012-May 2016

RELATED COURSEWORK

- Wireless Technologies and Design Concepts - CDMA, CDMA-2000, WCDMA, EV-DO, HSPA, LTE, LTE- Advanced, WiMAX (OFDMA based), DAS infrastructure, 5G Small cells, EDGE, GPRS, UMTS, WLAN, CBRS
- Technical 3GPP standards, VoLTE, 5G NR, NB-IoT. Protocols - RIP, UDP, TCP, IP, SIP
- Cellular Infrastructure - FTTH, BBU Processing, CPRI, Co-siting technique
- Network Design Concepts - C-RAN, Network Slicing, C/U Split, Network and Cell Virtualization, Fiber Front Haul
- MRF Planning and Antenna Design and effects of Massive MIMO and mm-Wave Domain

TECHNICAL SKILLS

- Tools - Matlab, Keil, Microsoft Excel, BlueBeam
- RF Design Tools and Software - iBwave, WindCatcher, DB browser, MapInfo, XCAP, XCAL
- Programming Languages - Python, Matlab, Database Languages - SQL, Scripting Environment - PowerShell, Stacks, Data Structure
- Advance Testing and Troubleshooting of Industry Accepted Tools - Anritsu, CommScope, EXFO
- Hands-on Experience - RF Plumbing Design and Testing of New Base-Stations

PROFESSIONAL EXPERIENCE

Skyline Communications, 12002 Beverly Park Rd., Everett, WA

08/08/2019 - Present

RF & Electronic Security Designer & Estimator

- Design & Optimization of Wireless Networks for in-building radio coverage (DAS Distributed Antenna Systems) for 5G, LTE, 4G, GSM, WCDMA, CDMA, CBRS, OFDM, NR, WIFI, WiGig and other wireless radio networks.
- Improvement of Cellular Infrastructure for 2 cities in US using FTTH, BBU Processing, RRH, CPRI protocol, C-RAN, Network Slicing, C/U Split, Network and Cell Virtualization, Fiber Front Haul by 18.6%.
- Design of Telecommunications networks with optical fiber & Cat6 cable; Integration of the networks with small cells for 5G technology.
- Provide indoor & outdoor RF Plumbing Design Planning. Perform & validate RF measurements & link budget for networks. Prepare KPI summary, Submittal Plans and Data Sheets included HO success rate, accessibility and retainability.
- Review prospective projects, summarize key points and bid strategy. Prepare material and labor estimates based on plans and specifications. Get competitive material quotes from vendors and select suppliers. Validate Co-siting Technique, Edge Computing and Evaluating Datasheets of Cell-Site Components.
- Prepare and submit proposals to selected customers. Provide consultative selling for private bid projects. Prepare comprehensive labor material budget for project handover to PM
- Review project for value engineering opportunities. Change order preparation and tracking. Maintain estimating files and supporting documentation

Teleworld Solution, 43130 Amberwood Plaza, Suite 210, South Riding, VA

06/06/2018 - 08/15/2018

RF Intern

- Performed RF measurements for coverage and capacity including traffic analysis, site zoning support evaluation and network optimization. Assisted field RF engineers by providing back office support, design and development in Sprint-VOLTE project.
- Validate Co-siting Technique, Edge Computing and Evaluating Datasheets of Cell-Site Components
- Carried out post processing of drive test data via XCAP and analyzed layer 3 messages for handovers and call drops in LTE using WindCatcher. Prepared test cases for drive test team based on the discrepancies.
- Identified sites with performance degradation issues. Improved coverage and efficiency to 40% by implementing macro design.
- Generated walk-test and drive-test routes via existing legacy for 5 outdoor sites using MapInfo.
- Enhanced signal strength (RSRP), RSRQ and signal quality (SINR) using 4G LTE of Kansas City.
- Generated walk-test and drive-test routes via existing legacy for outdoor sites using MapInfo.

Infosys Pvt. Ltd., Hebbel Electronics City, Mysore, KA, India

03/27/2017 - 06/16/2017

Systems Engineer

- Developed and implemented new software programs using programming languages: Python and C. Improved the quality of programs by adopting an object-oriented programming approach.
- Maintained and improved the performance of existing databases by 9% by applying Database Management System.
- Implemented programmable queries on Microsoft SQL Server 12 using database language SQL and MongoDB.
- Led a team of 4 in a project involving design of a real-time application software for booking of tickets & taxis, reservation in restaurant and hotel booking.

ITTM - A Unit of Mahanagar Telephone Nigam, Govt. of India, New Delhi

06/15/2015-07/15/2015

Telecom Industrial Trainee

Analyzed digital and analog components in a circuit. Participated in customer service handling along with the front desk team of MTNL and worked alongside with the installation team for deploying of sensors, detectors and routers. Completed research study on protocols: IP (IPv4 and IPv6), TCP, UDP. Configuration of components - DNS Server, DSLAM, BRAS, MDF and switches.

ACADEMIC PROJECTS & TECHNICAL PAPERS

- Scholarly Paper - DAS deployment solution for indoor LTE coverage Aug 2019-Dec 2019
 - DAS FloorPlanning using iBwave Sep 2018-Dec 2018
 - Cellular Network Design using Python Jan 2018-Apr 2018
 - Technical Paper - 4G technology Jan 2016-Apr 2016
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